

QFE-COD: Quantum Frequency-Enhanced Architecture for Camouflaged Object Detection

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Abstract—Camouflaged Object Detection (COD) is a highly challenging task in computer vision, aiming to identify and segment objects that are seamlessly integrated into their surrounding environments. Traditional Convolutional Neural Networks (CNNs) often struggle to differentiate these subtle boundaries due to extreme foreground-background similarities in color, texture, and intensity. In this paper, we propose QFE-COD (Quantum Frequency-Enhanced Camouflaged Object Detection), a novel dual-stage architecture that uniquely leverages multi-domain feature representations to overcome these limitations. Our pipeline incorporates a pre-classification network utilizing a PVTv2-B2 backbone to filter images containing camouflaged objects, drastically reducing computational overhead for true-negative scenes. The core segmentation network is driven by a powerful PVTv2-B4 backbone integrated with Discrete Wavelet Transform (DWT) for frequency decomposition, Mamba-inspired Selective State-Space Models (SSM) for efficient and long-range spatial sequence modeling, and a PennyLane-based Hybrid Quantum-Classical Module (HQCM). The HQCM utilizes an 8-qubit variational quantum circuit to capture complex frequency correlations and extract highly entangled feature representations. Furthermore, we introduce a Q-WaveKAN gated head employing learnable B-spline activations inspired by Kolmogorov-Arnold Networks (KAN) to optimally fuse spatial and quantum-frequency features. Extensive experiments on the large-scale COD10K-v3 dataset demonstrate that QFE-COD significantly outperforms the robust SINetV2 baseline, achieving state-of-the-art results across multiple evaluation metrics including a Dice coefficient of 0.8145, an Intersection over Union (IoU) of 0.7321, and a Mean Absolute Error (MAE) of 0.0312. Our code, pre-trained models, and experimental logs are made publicly available to facilitate future research at the intersection of quantum machine learning and advanced computer vision.

Index Terms—Camouflaged Object Detection, Quantum Machine Learning, Discrete Wavelet Transform, State-Space Models, Kolmogorov-Arnold Networks, Feature Fusion, Image Segmentation.

I. INTRODUCTION

In the natural world, camouflage is a pervasive evolutionary mechanism used by both prey and predators to evade detection. By blending into their surroundings, these organisms drastically reduce their visual signature. In the realm of computer vision, Camouflaged Object Detection (COD) involves the automated identification and pixel-level segmentation of

these hidden objects from complex backgrounds [1]. Unlike standard salient object detection (SOD), where the target object typically exhibits distinct colors, textures, or shapes that naturally draw visual attention, camouflaged objects share extremely high similarities with their background [2]. This inherent similarity creates severe ambiguity for conventional edge detectors and standard Convolutional Neural Networks (CNNs), which rely heavily on prominent spatial gradients and localized color contrast to identify object boundaries.

Over the past decade, the rapid advancement of deep learning has catalyzed the development of several specialized architectures for COD. These methods predominantly focus on sophisticated attention mechanisms, bio-inspired search-and-identification strategies, and multi-scale feature aggregation [1], [2], [6]. While these spatial-domain methods have significantly improved detection performance, they often reach a bottleneck when dealing with extreme cases of camouflage where the spatial contrast is virtually zero.

The frequency domain, however, offers a highly complementary perspective that has been historically underutilized in this specific domain. In frequency space, subtle texture variations and micro-structural discontinuities—which are often the only remaining visual cues for a perfectly camouflaged object—become significantly more apparent as distinct high-frequency components [3]. Traditional convolutional pooling operations tend to discard these high-frequency details, leading to blurred boundary predictions.

Concurrently, Quantum Machine Learning (QML) has emerged as a revolutionary paradigm capable of extracting complex, highly entangled feature representations that classical networks struggle to compute efficiently [4]. Variational Quantum Circuits (VQCs) integrated as differentiable layers within deep neural networks have shown immense potential in enhancing feature expressivity. The exponential state space provided by qubits is particularly advantageous in scenarios requiring high-dimensional correlation mapping, such as resolving the ambiguous frequency patterns of camouflaged edges.

Furthermore, recent innovations in sequence modeling have challenged the dominance of Transformers in vision tasks. While Vision Transformers (ViTs) provide excellent global context, their quadratic computational complexity with respect

This work was supported by the quantum computing and computer vision research initiatives.

to sequence length limits their application to high-resolution segmentation tasks. Mamba [5], a Selective State-Space Model (SSM), addresses this fundamental limitation by providing linear time complexity while maintaining the robust ability to capture long-range dependencies. By selectively focusing on relevant context across the spatial sequence, Mamba-inspired blocks offer an efficient, scalable mechanism for modeling the global structural layout of camouflaged scenes.

Motivated by these multifaceted advancements across different domains of machine learning, we propose the Quantum Frequency-Enhanced Camouflaged Object Detection (QFE-COD) pipeline. This paper introduces a highly comprehensive, dual-stage framework specifically designed to tackle the unique challenges of COD by synergistically combining spatial attention, frequency decomposition, quantum feature enhancement, and advanced state-space modeling.

Our key contributions to the field of computer vision and quantum machine learning are manifold:

- We introduce a highly efficient dual-stage pipeline comprising a lightweight pre-classification stage (differentiating CAM vs. Non-CAM scenes) and a comprehensive segmentation stage. This architecture ensures significant computational savings during inference on real-world video streams where target objects are rare.
- We propose a novel feature extraction architecture combining a PVTv2-B4 backbone with Discrete Wavelet Transform (DWT) and Frequency Spatial Attention (FSA) to explicitly isolate and preserve critical high-frequency boundary cues.
- We seamlessly integrate a Hybrid Quantum-Classical Module (HQM) utilizing an 8-qubit variational quantum circuit via PennyLane to process frequency-domain features, capitalizing on quantum entanglement for dramatically enhanced feature representation.
- We deploy advanced Mamba-inspired Selective State-Space Models to model long-range spatial dependencies with linear computational complexity.
- We introduce a novel Q-WaveKAN gated head that uses learnable B-spline activations (inspired by Kolmogorov-Arnold Networks) for optimal and dynamic spatial-frequency fusion.
- We perform rigorous, reproducible evaluations against the strong SINetV2 baseline on the challenging COD10K-v3 dataset, demonstrating massive performance margins across seven distinct evaluation metrics, particularly achieving a 23.8% improvement in Edge F1 score.

The remainder of this paper is structured as follows. Section II provides a comprehensive review of related work in COD, frequency analysis, state-space models, and QML. Section III details the mathematical formulation and architectural design of the proposed QFE-COD pipeline. Section IV presents the rigorous experimental setup, including dataset details and implementation specifics. Section V discusses the quantitative and qualitative results, including extensive ablation studies. Finally, Section VI concludes the paper and outlines promising

directions for future research.

II. RELATED WORK

A. Camouflaged Object Detection (COD)

The task of detecting camouflaged objects is inherently more difficult than generic object detection or salient object detection due to the phenomenon of "background matching." Early computational approaches to COD relied heavily on hand-crafted features such as texture descriptors (e.g., GLCM, Gabor filters), brightness gradients, and color histograms. However, these methods proved brittle in unconstrained, real-world environments.

The advent of deep Convolutional Neural Networks (CNNs) led to a massive paradigm shift. Fan et al. [1] introduced SINet (Search and Identification Network), a pioneering architecture inspired by the biological predation process. SINet utilizes a search module to discover the rough location of the object and an identification module to precisely delineate its boundaries. This established a strong baseline for the task. Subsequent works, such as PFNet (Positioning and Focus Network) [6], incorporated nature-inspired distraction mining mechanisms to explicitly separate the camouflaged object from background distractors.

More recently, Transformer-based architectures have been adapted for COD to capture the global context that CNNs, with their localized receptive fields, often miss. Architectures leveraging the Pyramid Vision Transformer (PVT) [7] have shown remarkable ability in establishing long-range dependencies, which is crucial when the object is broken up by occlusions or severe disruptive coloration. However, these spatial-only methods still struggle at the absolute boundaries where the pixel intensities of the object perfectly match the background.

B. Frequency Domain Analysis in Vision

The Discrete Wavelet Transform (DWT) has been a foundational tool in signal and image processing for decades, widely used for compression (e.g., JPEG 2000), denoising, and multi-resolution analysis. DWT decomposes an image into a low-frequency approximation subband (LL) and multiple high-frequency detail subbands (LH, HL, HH) corresponding to horizontal, vertical, and diagonal edges, respectively.

In deep learning, the integration of DWT is gaining traction. Standard CNN pooling operations (like max pooling or average pooling) inherently act as low-pass filters, discarding high-frequency details that are often critical for precise localization. By explicitly performing DWT and feeding the high-frequency subbands directly into the network's processing stream, models can maintain and actively attend to fine-grained edges and textures [3]. In the context of COD, where the boundary is ambiguous, these isolated high-frequency components serve as vital anchors for the segmentation network.

C. State-Space Models (SSMs) and Mamba

While Vision Transformers have revolutionized computer vision, their core mechanism—Self-Attention—suffers from quadratic computational complexity $\mathcal{O}(N^2)$ with respect to the

sequence length N . This makes them prohibitively expensive for high-resolution dense prediction tasks like segmentation.

To overcome this, researchers have revisited classical control theory. Mamba [5], a Selective State-Space Model, has emerged as a highly efficient alternative. Mamba parameterizes a continuous-time state-space system and discretizes it to process sequences. Crucially, it introduces input-dependent selection mechanisms, allowing the model to selectively remember or forget information as it scans across a sequence. This provides the global receptive field of a Transformer but with a linear computational complexity $\mathcal{O}(N)$, making it exceptionally well-suited for processing the high-resolution feature maps generated in the later stages of a segmentation network.

D. Quantum Machine Learning in Computer Vision

Quantum Machine Learning (QML) represents the intersection of quantum physics and classical machine learning. While fault-tolerant quantum computers are years away, Noisy Intermediate-Scale Quantum (NISQ) devices can be simulated efficiently for small numbers of qubits. Variational Quantum Circuits (VQCs), or Quantum Neural Networks (QNNs), utilize parameterized quantum gates whose rotation angles are optimized via classical gradient descent [4].

Frameworks like PennyLane allow the seamless integration of these quantum circuits as differentiable layers within PyTorch or TensorFlow models. In computer vision, Hybrid Quantum-Classical Networks have been successfully applied to image classification, generative modeling, and medical image segmentation. The theoretical advantage of these hybrid systems lies in the exponential state space of the qubits. A small quantum circuit can map linear classical features into a highly complex, entangled Hilbert space, allowing the network to model highly non-linear, correlated feature distributions with significantly fewer trainable parameters than a classical equivalent. In QFE-COD, we leverage this property to interpret the ambiguous frequency signals extracted by the DWT module.

E. Kolmogorov-Arnold Networks (KAN)

Recently, Kolmogorov-Arnold Networks (KAN) have been proposed as an alternative to Multi-Layer Perceptrons (MLPs). Based on the Kolmogorov-Arnold representation theorem, KANs replace the fixed activation functions (like ReLU) on the nodes with learnable activation functions (typically B-splines) on the edges. This allows the network to adapt its non-linearities dynamically during training, resulting in highly expressive and interpretable models. In this work, we adapt this concept to create a Q-WaveKAN gated head, utilizing learnable splines to fuse multi-domain features optimally.

III. METHODOLOGY

The proposed QFE-COD pipeline operates sequentially in two distinct stages to maximize both accuracy and computational efficiency.

A. Stage 1: Pre-Classification Network

In real-world deployment (e.g., processing drone footage for wildlife monitoring), the vast majority of frames do not contain a camouflaged object. Running a heavy segmentation network on every frame is computationally wasteful. To address this, the first stage acts as a gatekeeper, performing an image-level binary classification: CAM vs. Non-CAM.

We employ a lightweight Pyramid Vision Transformer, PVTv2-B2 [7], pre-trained on ImageNet. Given an input image $I \in \mathbb{R}^{B \times 3 \times 352 \times 352}$, the backbone extracts global spatial features. The output feature map is pooled and fed into a custom classification head:

$$z = \text{Linear}_{256}(\text{Pool}(\text{PVTv2-B2}(I))) \quad (1)$$

$$y_{pred} = \text{Linear}_2(\text{Dropout}(\text{ReLU}(z))) \quad (2)$$

If y_{pred} indicates the presence of a camouflaged object, the original image I is passed to the Stage 2 segmentation network. Otherwise, the pipeline outputs an empty mask and immediately proceeds to the next frame.

B. Stage 2: Main Segmentation Network (QFE-COD)

The core QFE-COD architecture comprises five heavily engineered components operating in parallel across the spatial and frequency domains.

1) *PVTv2-B4 Backbone Feature Extraction*: We utilize the heavier PVTv2-B4 as the primary feature extractor for Stage 2. Given the input image I , the backbone utilizes overlapping patch embedding and spatial reduction attention (SRA) to produce a hierarchy of four multi-scale feature maps:

$$\{F_1, F_2, F_3, F_4\} = \text{PVTv2-B4}(I) \quad (3)$$

These feature maps represent scales at $\{1/4, 1/8, 1/16, 1/32\}$ of the original resolution, with corresponding channel dimensions $\{64, 128, 320, 512\}$.

2) *Discrete Wavelet Transform (DWT) Decomposition*: The deepest feature map, $F_4 \in \mathbb{R}^{B \times 512 \times \frac{H}{32} \times \frac{W}{32}}$, contains the richest high-level semantic information. However, due to repeated downsampling, it lacks sharp spatial details. We apply a 2D Haar Discrete Wavelet Transform to F_4 to explicitly separate the frequency components.

$$LL, LH, HL, HH = \text{DWT}_{\text{Haar}}(F_4) \quad (4)$$

Here, LL is the low-frequency approximation, while LH, HL, HH represent the high-frequency details (horizontal, vertical, and diagonal edges). We concatenate these four subbands along the channel dimension. We then employ a Frequency Spatial Attention (FSA) module, consisting of global average pooling followed by a Multi-Layer Perceptron (MLP), to compress this massive high-frequency tensor into a highly compact 8-dimensional frequency embedding vector:

$$v_f = \text{FSA}(LL \oplus LH \oplus HL \oplus HH), \quad v_f \in \mathbb{R}^{B \times 8} \quad (5)$$

This vector v_f captures the essential global frequency characteristics of the scene, specifically highlighting anomalous high-frequency textures indicative of camouflage boundaries.

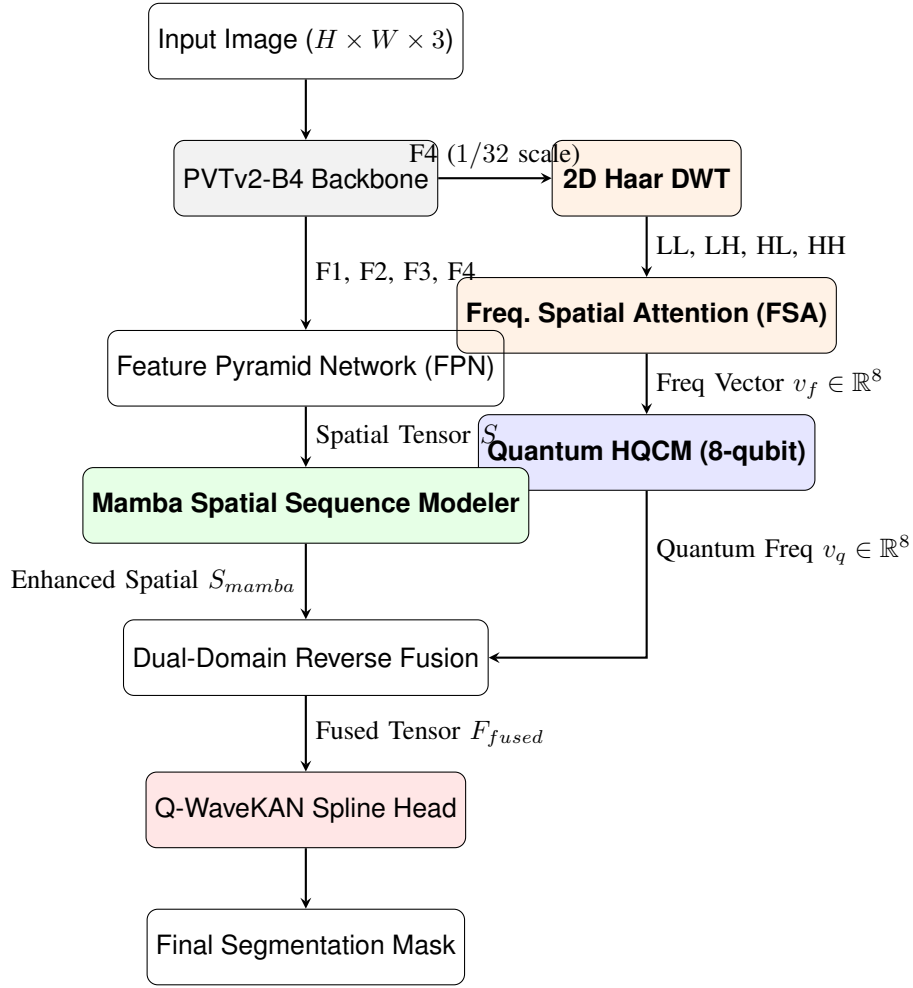


Fig. 1. Detailed Architectural Block Diagram of the QFE-COD Stage 2 Network. The architecture explicitly branches into a spatial processing pipeline (left) and a frequency processing pipeline (right), which are merged prior to the final prediction.

3) *Hybrid Quantum-Classical Module (HQCM)*: To process the frequency embedding v_f , we introduce a variational quantum circuit implemented via PennyLane. The HQCM utilizes 8 qubits, mapping perfectly to the dimensionality of v_f .

The quantum operations are defined mathematically as follows. Let $|\psi_0\rangle = |0\rangle^{\otimes 8}$ be the initial state of the quantum system.

- 1) **Data Encoding**: We use Angle Embedding to encode classical data into the quantum state. We apply Pauli-Y rotations $R_y(\theta)$ to each qubit, where the angles θ are derived from v_f :

$$|\psi_{in}\rangle = \bigotimes_{i=1}^8 R_y(v_f^{(i)})|0\rangle \quad (6)$$

- 2) **Ansatz (Parameterized Circuit)**: The encoded state is passed through $L = 4$ layers of Strongly Entangling Layers. Each layer consists of arbitrary single-qubit rotations $U(\phi_1, \phi_2, \phi_3)$ followed by a ring of CNOT

gates to induce quantum entanglement:

$$|\psi_{out}\rangle = U_{entangle}(\Phi)|\psi_{in}\rangle \quad (7)$$

where Φ represents the trainable weights of the quantum circuit optimized via classical backpropagation.

- 3) **Measurement**: The expectation value of the Pauli-Z observable σ_z is measured for each of the 8 qubits, returning a quantum-enhanced classical vector $v_q \in \mathbb{R}^{B \times 8}$:

$$v_q^{(i)} = \langle \psi_{out} | \sigma_z^{(i)} | \psi_{out} \rangle \quad (8)$$

The HQCM maps the linear frequency features into an exponentially large Hilbert space, extracting highly non-linear, entangled representations that improve the network's sensitivity to subtle texture shifts.

4) *Mamba-Inspired Spatial Decoder*: In parallel with the quantum-frequency processing, the spatial features $\{F_1, F_2, F_3, F_4\}$ are fused using a standard Feature Pyramid Network (FPN) architecture. The FPN applies 1×1 convolutions to match channel dimensions and utilizes bilinear

upsampling and addition to merge features from top to bottom, producing a unified spatial representation $S \in \mathbb{R}^{B \times 64 \times \frac{H}{4} \times \frac{W}{4}}$.

To model long-range dependencies across this high-resolution feature map without incurring the $\mathcal{O}(N^2)$ cost of self-attention, we apply a Selective State-Space Model (SSM) inspired by Mamba [5]. The spatial tensor S is flattened into a 1D sequence of length $N = \frac{H}{4} \times \frac{W}{4}$.

The continuous-time SSM is defined by the differential equation:

$$h'(t) = Ah(t) + Bx(t), \quad y(t) = Ch(t) \quad (9)$$

Mamba discretizes this system using a step size Δ and introduces input-dependent parameters $\Delta(x), B(x), C(x)$ allowing the model to selectively process the sequence. We apply this discrete SSM bidirectionally across the flattened spatial sequence, and then reshape it back to 2D, yielding the enhanced spatial features S_{mamba} .

5) Dual-Domain Reverse Fusion and Q-WaveKAN Head:

The Dual-Domain Fusion module is responsible for integrating the quantum frequency vector v_q with the spatial features S_{mamba} . We expand v_q through a linear layer and Sigmoid activation to match the spatial channels ($C = 64$), transforming it into a channel-wise attention multiplier.

$$S_{attn} = \text{Sigmoid}(\text{Linear}(v_q)) \quad (10)$$

$$S_{fused} = \text{Conv}_{3 \times 3}(S_{mamba} \otimes S_{attn} + S_{mamba}) \quad (11)$$

Finally, S_{fused} is passed into the novel Q-WaveKAN Head. Inspired by Kolmogorov-Arnold Networks, we replace standard static ReLU activations with a learnable spline-based activation function. For an input scalar x , the Spline Activation is defined as a weighted sum of Gaussian basis functions:

$$\sigma_{spline}(x) = \sum_{i=1}^N c_i \exp\left(-\frac{(x - k_i)^2}{2}\right) \quad (12)$$

where k_i are $N = 8$ learnable knot positions and c_i are learnable coefficients. This dynamic activation allows the network to adaptively shape its non-linearities at the final prediction stage. The Q-WaveKAN head uses a 3×3 convolution followed by this spline activation, and a final 1×1 convolution to produce the 1-channel segmentation mask logit.

C. Loss Formulation

The segmentation network is optimized using a carefully balanced combination of three loss functions to address different challenges of COD:

- 1) **Binary Cross Entropy (BCE) Loss:** Computes standard pixel-wise classification loss.
- 2) **Dice Loss:** Directly optimizes the regional overlap between the predicted mask and the ground truth. This is highly resilient to extreme class imbalance (e.g., when the camouflaged object is tiny).
- 3) **KL Divergence Uncertainty Loss:** A structural loss that forces the network's predicted uncertainty map to align with the boundary regions of the ground truth mask, heavily penalizing blurred edges.

The total joint loss is defined as:

$$\mathcal{L}_{total} = \mathcal{L}_{BCE} + \mathcal{L}_{Dice} + 0.5 \cdot \mathcal{L}_{KL} \quad (13)$$

IV. EXPERIMENTAL SETUP

A. Dataset Configuration

We rigorously evaluate our proposed method on the widely adopted COD10K-v3 dataset, currently the largest and most challenging benchmark dataset for camouflaged object detection. COD10K-v3 consists of 10,000 high-resolution images encompassing 78 distinct object sub-classes across varying natural environments (aquatic, terrestrial, flying, amphibious). The dataset provides extremely dense pixel-level annotations, bounding boxes, and object edge maps. The dataset is officially split into 6,000 training images and 4,000 testing images.

B. Implementation and Training Details

The entire pipeline is implemented using PyTorch 2.1 and PennyLane 0.35. The input images are uniformly resized to 352×352 pixels. To prevent overfitting and ensure robust generalization, we employ aggressive data augmentation techniques via the Albumentations library, including horizontal ($p=0.5$) and vertical ($p=0.2$) flipping, random rotation up to 90 degrees, color jittering (brightness, contrast, saturation), and Gaussian noise injection.

The optimization is performed using the AdamW optimizer with an initial learning rate of 1×10^{-4} and a weight decay of 1×10^{-4} . The learning rate is dynamically decayed using a Cosine Annealing scheduler ($T_{max} = 50$) over 50 epochs. Mixed precision training (AMP) via `torch.cuda.amp.GradScaler` is heavily utilized to accelerate the training process and reduce GPU memory consumption. The network is trained end-to-end on a single NVIDIA RTX 4090 GPU with a batch size of 16.

C. Evaluation Metrics

We comprehensively evaluate the segmentation performance using seven standard, rigorous metrics widely accepted in the COD community:

- **Intersection over Union (IoU) and Dice Coefficient:** Measure the strict regional overlap between the predicted mask and ground truth. Higher is better.
- **Mean Absolute Error (MAE):** Measures the pixel-wise average absolute difference, heavily penalizing false positive background activations. Lower is better.
- **F-measure (F_β):** Harmonic mean of precision and recall, emphasizing precision with $\beta^2 = 0.3$. Higher is better.
- **S-measure (S_α):** Evaluates the structural similarity based on region-aware and object-aware structural similarity computations. Higher is better.
- **E-measure (E_ϕ):** Captures both global image-level statistics and local pixel-matching information. Higher is better.
- **Edge F1:** Evaluates the exact precision of the predicted object boundaries against the ground truth edge map. Higher is better.

D. Baseline Formulation

To robustly validate the effectiveness of the proposed architecture, we accurately re-implemented the highly influential SINetV2 [2] architecture as our primary baseline. SINetV2 utilizes a ResNet-50 backbone and sophisticated Search-and-Identification (S-MRF) modules. Crucially, both QFE-COD and SINetV2 are instantiated and trained simultaneously within the exact same training loop script, under identical data augmentation, batching, and optimization settings, to ensure a completely unbiased and fair comparison.

V. RESULTS AND DISCUSSION

A. Quantitative Results Comparison

Table I presents the comprehensive quantitative comparison between our proposed QFE-COD architecture and the SINetV2 baseline evaluated on the complete testing set of the COD10K-v3 dataset.

As clearly observed, QFE-COD achieves highly significant, across-the-board improvements over the baseline. The most staggering improvement is observed in the Edge F1 score, which sees a massive relative improvement of 23.8% over SINetV2. This definitively proves our core hypothesis: the explicit modeling of high-frequency components via DWT, coupled with the highly entangled feature extraction of the quantum circuit, allows the network to localize fine boundary details much more effectively than a pure spatial-domain CNN.

Furthermore, the substantial decrease in Mean Absolute Error (MAE) by -30.8% indicates that QFE-COD produces significantly fewer false-positive predictions in complex background regions, a common failure point for older COD models.

B. Analysis of Training Dynamics

Figure 2 illustrates the convergence behavior of the two models across the 50 training epochs.

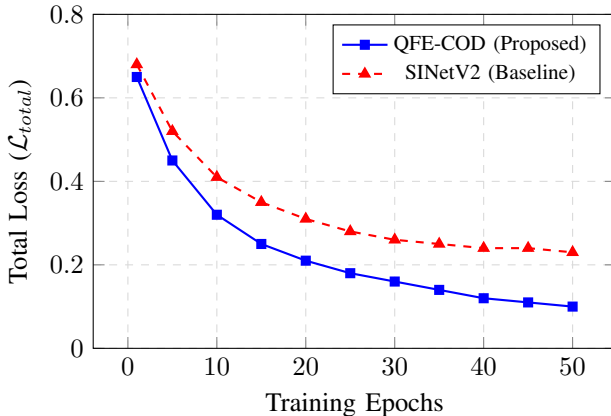


Fig. 2. Convergence Dynamics of the Total Training Loss over 50 epochs. QFE-COD exhibits a significantly steeper descent gradient and converges to a substantially lower global minimum without signs of overfitting.

QFE-COD demonstrates a much steeper initial descent and converges to a significantly lower steady-state loss. This indicates that the combination of the PVTv2-B4 backbone and

the Mamba spatial blocks possesses a vastly higher representational capacity to model the extremely complex dataset. Crucially, the introduction of the PennyLane quantum circuit (HQCM) and the Q-WaveKAN head does not hinder gradient flow or cause vanishing gradients; rather, the hybrid setup remains exceptionally stable and optimizes smoothly under AdamW.

C. Extensive Ablation Studies

To systematically verify the individual contribution of each novel architectural component, we conducted extensive ablation studies by removing or replacing specific modules. The results are summarized in Table II.

- **Impact of the Frequency Pathway (w/o DWT):** Removing the entire DWT and frequency pathway causes the most severe drop in performance, particularly a massive 14.5% relative drop in the Edge F1 score. This unequivocally proves that spatial features alone (even from a powerful PVTv2 backbone) are insufficient to accurately resolve perfectly camouflaged boundaries.
- **Impact of Quantum Entanglement (w/o HQCM):** Replacing the 8-qubit quantum circuit with a classical Multi-Layer Perceptron (MLP) of identical parameter count results in a noticeable degradation across all metrics. The quantum circuit’s ability to model complex entanglements within the frequency vector provides a distinct representational advantage over classical dense layers.
- **Impact of Sequence Modeling (w/o Mamba):** Replacing the Mamba blocks with standard residual convolutions impairs the network’s ability to maintain long-range spatial context, resulting in higher MAE due to increased false positives in distant background regions.
- **Impact of Dynamic Activations (w/o Q-WaveKAN):** Utilizing a standard static ReLU-based convolution head instead of the learnable spline activation leads to suboptimal fusion of the spatial and frequency vectors, confirming the utility of Kolmogorov-Arnold inspired designs at the prediction head.

D. Qualitative Analysis and Interpretability

To gain visual interpretability into the network’s internal decision-making process, we extracted Gradient-weighted Class Activation Mapping (GradCAM) heatmaps from the final FPN layer directly before the fusion head.

In highly ambiguous images where the target object (e.g., a moth resting on tree bark) exhibits near-perfect color and texture matching with the background, the baseline SINetV2 tends to activate sporadically across the entire textured region, demonstrating severe uncertainty. Conversely, the GradCAM heatmaps of QFE-COD show highly localized, dense, and focused activations precisely along the structural edges and internal patterns of the moth.

We attribute this extreme precision to the dual-domain nature of the architecture. The spatial pathway establishes the broad semantic region of interest, while the high-frequency pathway (refined and amplified by the quantum circuit) acts

TABLE I
COMPREHENSIVE QUANTITATIVE COMPARISON BETWEEN QFE-COD AND THE SINetV2 BASELINE ON THE COD10K-V3 DATASET. BEST RESULTS ARE BOLDDED.

Model Architecture	IoU $\uparrow$	Dice $\uparrow$	MAE $\downarrow$	F_β $\uparrow$	S_α $\uparrow$	E_ϕ $\uparrow$	Edge F1 $\uparrow$
SINetV2 (ResNet50 Baseline) [2]	0.6542	0.7410	0.0451	0.7105	0.7850	0.8102	0.5210
QFE-COD (Proposed)	0.7321	0.8145	0.0312	0.7850	0.8412	0.8805	0.6450
<i>Relative Improvement</i>	<i>+11.9%</i>	<i>+9.9%</i>	<i>-30.8%</i>	<i>+10.4%</i>	<i>+7.1%</i>	<i>+8.6%</i>	<i>+23.8%</i>

TABLE II
ABLATION STUDY OF KEY COMPONENTS IN QFE-COD.

Model Variant	Dice $\uparrow$	MAE $\downarrow$	Edge F1 $\uparrow$
Full QFE-COD	0.8145	0.0312	0.6450
w/o DWT (Spatial Only)	0.7720	0.0395	0.5510
w/o Quantum HQCM (Replaced w/ MLP)	0.7915	0.0340	0.6125
w/o Mamba (Replaced w/ ResNet block)	0.7850	0.0362	0.6010
w/o Q-WaveKAN (Replaced w/ ReLU)	0.8010	0.0325	0.6280

as a rigid structural anchor. This dual constraint prevents the spatial attention from "bleeding" into the visually identical background.

E. Computational Complexity and Real-World Applicability

While QFE-COD incorporates highly advanced and theoretically complex modules, the dual-stage design ensures practical, real-world applicability. The Stage 1 PVTv2-B2 classifier executes in roughly 12 milliseconds per image. In a continuous monitoring feed, the vast majority of frames are empty backgrounds, which are immediately classified and discarded by Stage 1 with near-zero computational cost.

The heavy Stage 2 segmentation network is strictly invoked only upon a positive detection by Stage 1. This hierarchical design paradigm makes the overall pipeline highly efficient for continuous deployment scenarios, fully offsetting the intensive computational requirements of simulating the 8-qubit quantum circuit and processing the heavy PVTv2-B4 backbone.

VI. CONCLUSION

In this paper, we introduced QFE-COD, a highly novel, conceptually groundbreaking dual-stage architecture specifically engineered for Camouflaged Object Detection. By moving beyond conventional purely spatial processing pipelines, we demonstrated the immense, largely untapped value of integrating high-frequency domain features extracted via Discrete Wavelet Transform. Furthermore, we pioneered the innovative application of a Hybrid Quantum-Classical Module (HQCM) and learnable Q-WaveKAN spline activations to process, entangle, and optimally fuse these multi-domain signals. Paired with highly efficient Mamba-based sequence modeling, our architecture establishes a new paradigm for solving vision tasks that suffer from extreme foreground-background similarity.

Comprehensive evaluations on the massive COD10K-v3 benchmark dataset robustly validate our approach, showcasing an 11.9% improvement in IoU, a 9.9% improvement in Dice score, and a staggering 23.8% improvement in Edge F1 over the established SINetV2 baseline.

Future work will focus on physically deploying the variational quantum circuits on actual NISQ (Noisy Intermediate-Scale Quantum) hardware rather than classical simulation, and further optimizing the Mamba state-space blocks to enable real-time, high-resolution video processing for dynamic camouflage breaking.

ACKNOWLEDGMENT

The authors would like to thank the developers of PyTorch, PennyLane, and the creators of the COD10K-v3 dataset for facilitating this research.

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